Adam Rida

Data Scientist and AI Research Scientist
France & USA — +33 (0) 6 51 09 99 93

Portfolio: adrida.github.io
Email: adamrida.ra@gmail.com
Github – Google Scholar

Experience

Autoplay AI — Paris / New York City
Head of AI - Founding Team — *Jan 2025 – Present*

- Architected a real-time inference pipeline ingesting millions of interaction events from thousands of session replays, generating behavioral scores in milliseconds
- Shipped a hybrid LLM + ML model suite that detects hesitation, knowledge gaps and intent, taking the project from concept to production in < 3 months
- Surfaced "struggling-user" cohorts through unsupervised clustering, driving targeted UX fixes with human-readable insight reports
- Indexed multi-million embeddings in a Pinecone vector store for sub-second semantic search across all user profiles
- Conceived and prototyped TERRA - a Task-Event Representation & Reasoning Architecture for automated large-scale session-replay analysis; conference manuscript in preparation

Entrepreneur First - SynthAI — Paris, France
Co-founder and CTO — *May 2024 – Dec 2024*

- Developed an AI co-pilot for investment thesis generation in private equity, releasing three live prototypes to over 100 private equity funds
- Engineered a proprietary PDF parser and innovative multi-document summarization techniques using knowledge graphs (GraphRAG and Multi-headed RAG)
- Led product design, development, and AI research, addressing RAG limitations in summarization
- Acquired by one of our customer

MFG Labs — Paris, France
Data Scientist II — *Nov 2023 – Apr 2024*

- Optimized supply chain operations for a luxury brand using AI, tackling data scarcity challenges
- Designed strategies to minimize stockouts while effectively meeting customer demand

AXA / Sorbonne University – CNRS — Paris, France
AI Research Scientist – Ph.D. Candidate — *Oct 2022 – Oct 2023*

- Conducted research on Responsible and Explainable AI within AXA's R&D team
- Developed new approaches for differential explanations to interpret differences between models
- Authored a research paper accepted at ECML/PKDD 2023 workshop

Société Générale CIB — Paris, France
Data Scientist — *Sept 2021 – Sept 2022*

- Created predictive analytics tools and dashboards for sales teams, boosting performance significantly
- Processed large datasets (>20M rows monthly) using PySpark and Kedro.
- Implemented code guidelines and team management processes for a 10+ team.

Rebellion Research — NYC, USA
Machine Learning Quantitative Researcher Intern — *Feb 2021 – Sept 2021*

- Constructed deep reinforcement learning investment strategies, achieving a 21% annualized return and 1.8 Sharpe Ratio
- Engineered features from SEC 13-F filings to enhance model performance
- Published essays on AI and cryptocurrency, reaching over 100,000 readers

Qantev — Paris, France
Data Scientist Intern & Software Engineer — *Apr 2021 – Oct 2021*

- Achieved 97% accuracy in medical insurance NLP models, up from 70%
- Optimized data processing architecture, increasing speed by 70%.
- Collaborated in an agile, test-driven environment with high code standards.

Education

Sorbonne University – CNRS - LIP6 – AXA Paris, France
Ph.D. Candidate in AI *2022 – 2023*

- Specialized in Explainable AI (XAI) focusing on differential explanations between models.
- Part of the AXA-Sorbonne University Trustworthy and Responsible AI Lab (TRAIL), supervised by Christophe Marsala, Marie-Jeanne Lesot and Xavier Renard.
- Transitioned to industry after one year because of a financial emergency.

CY Cergy Paris Université - CY Tech (ex EISTI - École d'Ingénieur) Cergy, France
Master's degree in Applied Mathematics and Data Science – (Diplôme d'Ingénieur) *2017 – 2022*

- Ranked in the top 8% with a grade of 16/20 (Grade A).

Ecole 42 Paris, France
Peer-to-peer Programming Program *2017 – 2020*

- Ranked 24th out of 900 applicants in a competitive one-month programming session

Lycée Charlemagne Paris 4, France
Baccalauréat (Honors) *2016*

- Among top Parisians high schools - French Baccalauréat with Honors, Mention Bien

Skills

- **AI & ML**: XAI, ML Model Dynamics, ML and Deep Learning, RAG pipelines, GraphRAG, Concept Drift, DRL, Transformers, BERT, NLP, GNNs, Summarization
- **Programming**: Python (most proficient language), R, Java, C, Scala, SQL
- **Tools & Frameworks**: Scikit-learn, pandas, numpy, PyTorch, TensorFlow, Kedro, PySpark, LangChain, Pinecone, Gradio
- **Languages**: French: Native — English: Fluent (TOEIC 965) — Spanish: Conversational

Publications & Projects *(all links, more details and access to code at adrida.github.io)*

- **Dynamic Interpretability for Model Comparison via Decision Rules**, Adam Rida, Marie-Jeanne Lesot, Xavier Renard, Christophe Marsala; ECML/PKDD 2023 DynXAI Explainable Artificial Intelligence: From Static to Dynamic Workshop
- **Deep Reinforcement Learning & Feature Extraction For Constructing Alpha Generating Equity Portfolios**, Alexander Fleiss*, Amrith Kumaar*, Adam Rida*, Junsup Shin, Xinying Lai, Vivian Fang, Jialiang Chen, Ang Li:: SSRN (800+ downloads *:equal contributors)
- **Temporal Augmented Retrieval (TAR) - Dynamic RAG** (Open Source Project - 27+ stars on Github)
- Essays on Rebellion Research: XAI, Crypto, Human–AI Coordination.
- Web3: Projects with 3K+ users and multi-million market caps (details on request).